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Operational Circularity and Industrial Demand Response Optimization in Renewable-Integrated Electric Arc Furnace Steel Production

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Abstract

This study develops an integrated techno-economic optimization framework for electricity-intensive Electric Arc Furnace (EAF)-based steel manufacturing systems operating under dynamic electricity-market conditions and industrial demand-response environments. The proposed methodology combines industrial production scheduling, photovoltaic self-generation, battery energy storage, thermal energy recovery, and operational circularity assessment within a unified Mixed-Integer Linear Programming (MILP) framework. The study additionally investigates the industrial electricity demand function under variable electricity-price conditions, demonstrating that steel-industry demand remains largely technologically constrained and responds primarily through temporal load shifting and operational scheduling rather than conventional elastic consumption behavior. The framework is applied to repeated 24-hour industrial production cycles over a rolling weekly horizon under three representative electricity-market scenarios. Results indicate that operational flexibility, renewable-energy integration, and thermal buffering substantially reduce operational costs, improve renewable self-consumption, enhance waste-heat utilization, and lower emissions intensity while preserving metallurgical stability and minimizing scrap generation. The study further introduces the concept of operational circularity, extending circular-economy approaches beyond material recycling toward integrated renewable-energy utilization, thermal recirculation, and quality-preserving industrial flexibility. The findings highlight the strategic role of industrial demand response and sector-coupled optimization in supporting low-carbon industrial transition pathways.

Keywords: Industrial Demand Response; Operational Circularity; electricity -market dynamics; Circularity Performance Index;

1. Introduction

The decarbonization of the industrial processes has become one crucial challenge into the climate neutrality and the energy transition phase of the upcoming years. There are several industrial sub-sectors facing this challenge, nevertheless the steel industry holds a significant role, thanks to its high energy intensity and the remarkable contribution to the global greenhouse gas emissions (GHG). When the Steel production is based on the Electric Arc Furnace (EAF) process, then the main consumption carrier is the electricity instead of the solid fuel and represents a reliable and countable pathway to achieve a low-carbon steel

manufacturing, especially when combined with the integration of renewable resources and the storage of the renewable energy generation. However, the introduction of the EAF design to the steel manufacturing increase the exposure of the entire process to the modern electricity markets, including the risks and challenges that are relevant to the electricity price volatility, electrical system flexibility and the optimization of the industrial operation for steel production. The variability of the electricity markets in Europe and Greece is a strong temporal characteristic in European and especially in Greek market level, since it exhibits temporal variations of shares in renewable sources or natural gas-driven generation. The interaction between renewable generation, which is not stable during intraday time, the quantities of inelastic demand and the contribution of gas-driven plants is essential for the formation of the core pattern of pricing. Therefore, there are noticeable intraday price spreads, corresponding to either high renewable shares and low-price time slices or gas-fired generation with high prices periods. These dynamics generate both risk and opportunities for the end users of the industrial sector.

Within the above context, the Demand Response (DR) function becomes a strategic mechanism for industrial electricity consumers, therefore industrial installations such as steel plants which are based on the EAF process are capable to actively respond to the above-mentioned price signals. So, the organizers of these plants prefer to shift, adjust and sequence their process steps, and maximize their cost avoidances during the appropriate price time slices. It is about a remarkable transformation that is going to turn the industrial loads to flexibility assets, in an absolutely effective pathway and to balance the entire load system while the operating costs are reduced at the same time.

The current research gives a structured optimization framework for an EAF-based steel plant, incorporating the process sequence, electricity price signals, renewable generation with Photovoltaics (PVs), battery storage, thermal energy recovery. This model is based on a hourly time resolution, capturing the interaction between production stages and the flexibility of the entire energy system covering the industrial plant. More specifically, the framework draws a certainly defined technical sequence of production stages, including EAF melting, secondary downstream processing, thermal buffering, and preparation phase, and at the same time the intraday placement of the stages is optimized within each individual 24 – hour operation cycle. The main innovation of this study is based on the combination of the scheduling of the manufacturing process with energy management optimization, following the real electricity and fuels markets conditions. The energy consumption is not treated as an exogenous procedure, but the model is explicitly embedding the production steps decision making into the optimization issue, targeting to the minimization of the entire operation cost, taking into account the electricity trading, the natural gas consumption, production inefficiencies derived by the transition of process stages, which are represented by the scrap rates. The adoption of battery storage and thermal buffering is further supporting the flexibility and the inter-temporal shifting of energy use throughout the intraday.

It is mentioned that the framework is extended beyond the single-day optimization, including multiple consecutive operation cycles, a 7 days period, corresponding to a normal production scheduling. At the same time, daily cycle constraints for state of Charge -SOC (both battery and thermal are maintained, reflecting the real-world industrial constraints, while the operations are continuous in a 24/7 mode, keeping stability across cycles.

The integration of the industrial Demand Response with process – level modelling, helps this study to contribute into bridging the gap industrial process engineering and energy economics. Therefore, a practical and sectoral policy addicted approach is provided, to evaluate the industrial flexibility, the cost efficiency and the energy transition targets and ambitions.

2. Theoretical Framework

2.1 Industrial Demand Response in Energy-Intensive Industries

Demand Response in industrial processes applies if the end users of a steel plant (for instance) are capable to adjust and modulate their consumption in electricity in response to the market signals of the Electricity prices, according to the spot prices level of Energy Exchange. At the same time they must be able to utilize their manufacturing process to produce their final products. Steel production industrial plants DR is extremely valuable due to the large and controllable loads included. In this case production planning drives the industrial DR, in contrast to the residential DR case, in which the consumption behavior of the end user is not configured in a uniform pattern of intraday scheduling. Therefore, industrial DR present such a suitable time scheduling to achieve the appropriate load shifts that are matching with the price signals. Steel manufacturing that is based on EAF technology present an electricity consumption profile that is concentrated in specific stages, such as the melting phase. Then, the process stages that are corresponding to high consumptions get the chance to be shifted into low prices time windows. However, there are process interdependencies, peculiar needs to ensure the production continuity between stages, product quality considerations (i.e. regarding scrap minimizing). These factors act as significant constraints to flexibility windows.

Therefore, the foundation of the industrial DR can be framed as a **cost minimization problem under operational constraints**, where the objective is to minimize the electricity, energy and production costs, considering the process sequence and the technical constraints derived by the needs of the manufacturing process.

2.2 Process Sequencing and Time Allocation

The manufacturing process is presented as a model with a sequence of separated stages, each one of them having a standard time duration and specific requirements in electricity, fueling etc. Since each process stage is assigned to a certain time slot, the technical order of the process is preserved. Therefore, the process steps for steel production consist of: a) EAF melting stages, b) secondary downstream processing, c) critical load stages, d) preparation for EAF stages. There are precedence constraints, to ensure that process stages will follow a specific order within the 24 – hour cycle. The starting time of the process sequence becomes the driving decision variable to determine the alignment of the entire process with the profile of the electricity prices dataset.

Then, cost minimization objective, energy balance and flexibility mechanisms, scrap rate and process efficiency have to be thoroughly considered.

In addition, the model has to include operational constraints such as the decisions for the battery and thermal recovery operations, the limits on charging or discharging actions per time slice (hourly) for battery and for discharging the thermal discharge actions and of course the constraints in process sequence. IF the above-mentioned constraints are included, then planning and scheduling of next 24-hour cycles will be realistic and avoid procedures that are physically infeasible during the operation.

2.3 Multi-day optimization and circular operation

The model is preferable to be applied over a 7-days cycle, and each day represents a full production cycle. The process is under continuity, thanks to the circular concept of the process. Each 24 – hour cycle is independent and at the same tile is linked throughout SOC constraints. Optimization concept is repeated for each cycle to ensure the consistency of the process. It is preferable to follow a 7 days cycle, since production planning is good to have a schedule over multiple days ahead, but not so many, in order to be updated and revised after potential disruptive event or a market signal reverse that is possible to take place in any upcoming period.

It is necessary for the framework to consider the market signals and incorporate them in an explicit way, since the electricity prices reflect the marginal cost and electricity market conditions, the natural gas prices represent the cost of thermal energy and the share of renewable power reflects the solar availability and the actual renewable sources contribution. Therefore, the model can simulate the realistic behavior of production planning, that is driven by the energy markets. Then, the model can evaluate the economic benefits of the flexibility and the DR functionality in steel production.

To conclude, the entire framework achieves a combination of the industrial steel production and process modelling, the optimization of energy use, and, at the same time presents the decision – making tool that is based on the energy markets. The framework demonstrates how the steel production based on EAF operation enables the flexibility demand resources, and how it contributes to the energy efficiency, cost optimization and the stability of the entire energy and production system.

3. Literature review and the contribution of the current study

3.1 Current literature on steel decarbonization and circularity

Recent research has extensively examined the environmental footprint, circularity potential, and decarbonization pathways of the steel industry. For example, the analysis of European industrial decarbonization pathways (Martinez 2024) highlights that energy-intensive industries such as steel, cement, and chemicals are central to achieving EU climate neutrality targets by 2050. However, these sectors still face major barriers in adopting low-carbon technologies due to cost, regulatory uncertainty, and infrastructure constraints.

Other studies focus on **technological improvements and digitalization** in steel manufacturing systems. For instance, (Stavropoulos 2022) proposes a framework for reducing CO₂ emissions in manufacturing through energy-efficient equipment and digital monitoring tools, demonstrating that operational improvements can reduce emissions significantly while maintaining production efficiency.

From the perspective of circular economy, research has investigated **resource recovery and recycling technologies** in steel production. (Simoni, et al. 2024) analyze the potential for recovering valuable materials from steelmaking residues such as dusts, slags, and sludges, emphasizing that improving waste recovery and recycling technologies is critical for reducing environmental impacts.

Similarly, studies on circular economy in the metallurgical sector highlight that steel is inherently well suited to circularity because of its high recyclability and the potential for closed-loop material cycles (Fichera, et al. 2020). However, the implementation of circular production systems requires improved management of energy flows, by-products, and industrial symbiosis between sectors.

Furthermore, the concept of industrial symbiosis has been proposed as a pathway to improve resource efficiency and energy utilization across industrial clusters. European process industries are exploring cross-sector energy and material exchanges to reduce emissions and enhance competitiveness (Branca, et al. 2024).

Finally, literature on industrial demand response shows that industrial processes can provide significant flexibility to the electricity system through load shifting, curtailment, or energy storage.

Nevertheless, significant research gaps remain and participation of heavy industries in demand response programs remains limited due to operational constraints and the critical nature of industrial loads. Most existing studies examine either emissions reduction, industrial circularity, renewable integration, or industrial flexibility separately rather than within an integrated operational framework. In particular, **limited** research combines **industrial process scheduling, electricity-market dynamics**, renewable-energy integration, battery storage,

1
2
3
4 thermal-energy recovery, and production-quality preservation in a unified optimization
5 environment.

6 Another important limitation concerns the representation of chronological industrial operation.
7 Existing studies frequently simplify industrial scheduling **by assuming fixed daily**
8 **boundaries or static operational behavior.** In practice, however, steel-production systems
9 operate continuously, with strong interdependencies between EAF melting stages, downstream
10 rolling processes, thermal continuity, and energy-storage utilization. Consequently, industrial
11 flexibility cannot be treated as unconstrained electricity-load shifting but must **preserve**
12 **process continuity and metallurgical stability.**

13 Finally, although circular-economy studies in steel production primarily focus on material
14 recycling, slag reuse, and by-product management, considerably less attention has been given
15 to **operational circularity**, including renewable-energy recirculation, thermal-energy
16 recovery, and integrated industrial-energy scheduling. Therefore, a need remains for integrated
17 techno-economic frameworks capable of simultaneously evaluating industrial demand
18 response, renewable integration, thermal recovery, operational circularity, and low-carbon
19 process optimization within modern EAF-based steel production systems.
20
21

22 3.2 Current Research Contribution

23 By optimizing repeated 24-hour production circles across a rolling weekly horizon, the
24 framework provides an economically optimal operational schedule for electricity-intensive
25 industrial facilities operating under modern energy transition conditions and demand response
26 environments.
27

28 3.2.1 Integrated Techno-Economic Decarbonization Perspective

29 The proposed optimization framework extends beyond conventional industrial production
30 scheduling and represents an integrated techno-economic energy management system for
31 electricity-intensive steel production operating under dynamic low-carbon electricity markets.
32 The methodology combines industrial flexibility, renewable-energy integration, thermal-
33 energy recovery, battery-storage coordination, and chronological process scheduling within a
34 unified decarbonization framework.

35 The architecture reflects sector-coupling principles, where industrial facilities interact
36 dynamically with electricity markets, renewable generation, thermal systems, and energy-
37 storage technologies. Consequently, the industrial plant is modeled not as a passive electricity
38 consumer, but as an active flexible energy participant capable of adapting production schedules
39 according to market conditions and renewable-energy availability. The optimization
40 simultaneously co-optimizes industrial electricity demand, photovoltaic self-generation,
41 battery-storage operation, thermal recovery, and chronological production scheduling,
42 capturing the interaction between industrial operation and renewable-rich electricity systems
43 characterized by variability and price volatility.
44

45 3.2.2 Sector Coupling and Industrial Energy Integration

46 The framework represents a sector-coupled industrial-energy system integrating electricity,
47 thermal energy, renewable generation, and industrial production processes within a common
48 optimization structure. The production line interacts directly with wholesale electricity
49 markets, natural-gas consumption, photovoltaic generation, battery-storage systems, and
50 industrial waste-heat recovery.

51 Electricity-intensive stages, particularly EAF operations, are strategically shifted toward low-
52 price and high-renewable periods, while thermal recovery systems reduce fossil-fuel demand
53 and battery storage enables temporal electricity-price arbitrage. Consequently, the optimization
54 problem becomes a coupled industrial-energy scheduling problem rather than a standalone
55 production-planning exercise.
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3.2.3 Renewable Integration and Industrial Flexibility

A central objective of the framework is the maximization of industrial flexibility under renewable-rich electricity-market conditions. Increasing renewable-energy penetration introduces strong hourly variability in electricity prices due to fluctuations in solar and wind generation. Industrial flexibility therefore becomes a valuable mechanism for shifting electricity demand toward renewable-rich and low-price periods while reducing exposure to fossil-fuel-driven price spikes.

The methodology introduces chronological process scheduling as a flexibility mechanism by allowing each production circle to begin dynamically at the economically optimal hour instead of operating within rigid calendar-day boundaries. This enables synchronization between electricity-intensive industrial stages and favorable market conditions.

Battery storage further enhances flexibility by storing electricity during low-price or high-solar periods and discharging during expensive market conditions. Similarly, the thermal-buffer subsystem stores recovered industrial heat and later reuses it in downstream operations, reducing natural-gas dependence and improving industrial energy efficiency.

3.2.4 CO₂ Emissions and Decarbonization Framework

In addition to operational cost minimization, the proposed framework contributes directly to industrial decarbonization objectives.

The optimization indirectly reduces carbon emissions through reduced grid electricity imports, increased renewable self-consumption, reduced natural gas demand, improved thermal efficiency, and chronological demand shifting away from carbon-intensive peak periods.

The total indirect CO₂ emissions associated with industrial operation are calculated based on hourly grid electricity imports and natural gas consumption using the corresponding electricity and fuel emission factors.

Because electricity emission intensity often varies hourly according to generation mix composition, shifting industrial electricity demand toward renewable-rich periods may simultaneously reduce both operating costs and indirect emissions.

Photovoltaic generation directly offsets grid electricity consumption, while battery discharge reduces dependence on electricity imports during fossil-fuel-dominated peak-price hours. Thermal recovery systems further contribute to decarbonization by reducing fossil thermal demand and improving industrial heat utilization efficiency.

Consequently, the proposed optimization framework supports industrial transition toward low-carbon manufacturing systems consistent with modern sustainability and energy transition objectives.

3.2.5 Sustainability Interpretation

The framework demonstrates how industrial facilities may evolve from passive electricity consumers into actively optimized low-carbon industrial-energy systems. The methodology combines industrial demand response, renewable-energy integration, battery-storage operation, thermal recovery, chronological scheduling flexibility, and operational decarbonization within a unified optimization architecture.

The resulting structure improves operational efficiency, renewable-energy utilization, industrial flexibility, thermal efficiency, and indirect emissions performance. Although the present implementation focuses on electricity, photovoltaic generation, storage integration, and thermal recovery, the framework remains compatible with future industrial decarbonization pathways including green hydrogen, ammonia-based heating, synthetic fuels, and advanced energy-storage systems.

Consequently, the methodology provides a scalable techno-economic optimization framework suitable for modern sustainable industrial systems operating under dynamic low-carbon electricity markets.

3.2.6 Operational Circularity

A major contribution of the present study is the **introduction of operational circularity** within steel-sector industrial prosumer systems. Existing circular-economy approaches in steel production mainly focus on recycling, slag valorization, scrap reuse, and by-product recovery. In contrast, this framework extends circularity beyond material flows and introduces a dynamic operational perspective in which **circularity emerges from the interaction among:**

- production scheduling,
- renewable-energy integration,
- thermal-energy reuse, and
- process-quality preservation.

The proposed concept of “circular scheduling” aligns industrial production with photovoltaic availability, low-carbon electricity periods, and waste-heat recovery opportunities, maximizing internal renewable-energy utilization while minimizing thermal losses and grid dependence. Furthermore, the framework introduces “quality-preserving curtailment,” where demand-response actions are constrained to avoid excessive thermal instability, abrupt ramping, and increased scrap generation, recognizing scrap as an operational failure of circularity due to irreversible material loss.

By integrating renewable self-consumption, waste-heat utilization, flexible production scheduling, and scrap minimization into a unified optimization framework, the study bridges an important gap between industrial demand-response literature and circular-economy research. Consequently, industrial circularity is reinterpreted as an operational optimization problem strongly linked to flexibility, process stability, and decarbonization under renewable-rich electricity systems.

4. Methodology

4.1 Optimization Framework Overview for Weekly Industrial Process-Circle Scheduling

The proposed methodology develops an integrated industrial demand response optimization framework for electricity-intensive mini-mill steel production systems operating under dynamic electricity market conditions. The framework combines industrial production scheduling, electricity market optimization, photovoltaic (PV) self-generation, battery energy storage systems, thermal recovery, and thermal buffer management into a unified short-term operational planning model.

The methodology is designed specifically for repeated 24-hour industrial production circles operating over a rolling working-week horizon. The objective is not long-term production planning, but rather the determination of the economically optimal short-term operational schedule capable of minimizing total production cost under hourly variations in electricity prices, natural gas prices, and renewable energy availability.

Unlike conventional daily optimization approaches, the proposed framework distinguishes explicitly between calendar days and industrial process circles. Calendar days represent the candidate market horizon selected from the hourly dataset, whereas process circles represent the actual industrial production loops optimized by the model. This distinction is essential because industrial production systems operate continuously and are not naturally constrained by midnight boundaries.

Therefore, each 24-hour process circle may begin at any hour within the selected calendar day and continue chronologically into the subsequent calendar day if necessary.

4.2 Definition of the Chronological Industrial Process Circles

- The industrial mini-mill process is modeled as a fixed chronological sequence of interconnected production stages characterized by specific electricity loads, thermal requirements, and operational dependencies. The production sequence includes:
 - Step 1: 4 hours, EAF1.1–EAF1.4
 - Step 2: 3 hours, Secondary Downstream 1.1–1.3
 - Step 3: 1 hour, Shut Down sequence
 - Step 4: 6 hours, Minimum Critical Load 1–6
 - Step 5: 3 hours, Preparation phase 1–3
 - Step 6: 4 hours, EAF2.1–EAF2.4
 - Step 7: 3 hours, Secondary Downstream 2.1–2.3

The decision problem is to determine the **start hour of Step 1**, denoted by the hour assigned to EAF1.1, so that the **total electricity cost** over the full 24-stage production cycle is minimized.

Because the sequence may begin late in the day, the remaining process stages continue into the **next 24-hour day**. Thus, the production horizon is represented on a **48-hour time axis**, although only **24 consecutive hours** are used for the selected cycle.

The mini-mill production cycle is modeled as a fixed 24-hour ordered process sequence. Let $t=1$ up to $t=24$ denote the relative process-time index, where $t=1$ corresponds to EAF1.1 and $t=24$ corresponds to Secondary Downstream 2.3. The decision variable selects the start hour h of the first day at which EAF1.1 begins

4.3 Core Technical and Operational Assumptions of the Industrial Mini-Mill Optimization Framework

The developed industrial demand response and process-optimization framework represents a modern scrap-based Electric Arc Furnace (EAF) mini-mill operating under high electrification and renewable-integration conditions. The model assumes an annual steel throughput of approximately 650,000 t/y, consistent with EAF configurations reported by the (World Steel Association 2021) and the (Joint Research Center (JRC) 2012) studies. It integrates photovoltaic generation (12 km²), battery storage (20 MWh), thermal-buffer recovery (30 MWh_{th}), and process flexibility within a unified optimization environment. Battery assumptions follow the (International Energy Agency-IEA 2020) study, while waste-heat recovery assumptions are supported by the (Mc Brien, Serrenho and Allwood 2016), (Schneider and Lechtebohmer 2018) studies, and reheating-furnace literature. Battery and thermal charge/discharge coefficients (R_b , R_{th}) of 0.14 approximate feasible flexible energy flows without violating operational continuity. Baseline scrap rate is set at 2.1%, with scrap cost at 650 €/t, based on (World Steel Association 2021), (Joint Research Center (JRC) 2012). (Castro, Sun and Harjunkoski 2013), and (Zhang, et al. 2015). Collectively, these assumptions establish a scientifically grounded techno-economic basis for evaluating industrial flexibility, renewable integration, waste heat recovery, circular energy utilization, and cost optimization strategies in electricity-intensive steel manufacturing systems. The core technical and operational assumptions are presented in Table 1.

Table 1. Core Technical and Operational assumptions

Core Technical and Operational assumptions

Parameter	Size	Unit	Meaning
PV area	12	Km2	on-site PV installed area
Battery energy	20	MWh	Usable battery capacity
Battery efficiency	0,92	fraction	One-way proxy in solver sheet
Thermal buffer capacity	30	MWh_th	Recoverable heat stored for later use
Battery Charge / Discharge Rate Rb (fraction)	0,14	fraction	Share of process load recoverable as useful heat
Thermal Charge / Discharge Rate Rth (fraction)	0,14	fraction	Additional coefficient for downstream / finishing heat
Baseline scrap rate	0,021	fraction	Reference yield loss before optimization
Scrap cost	650	€/t scrap	Lost value + rework proxy
Illustrative annual throughput	650000	t/y	Used only in report narrative

4.4 Electrical Loads Calibration

The electrical-load decomposition represents a scrap-based Electric Arc Furnace (EAF) mini-mill with nominal EAF demand of approximately 60 MW during melting periods and a constant critical baseload of 12 MW corresponding to cooling systems, pumps, compressed air, ventilation, and automation infrastructure. Additional downstream loads vary between 3–26 MW, resulting in total plant demand between 12–80 MW and annual electricity intensity close to 517 kWh/t steel. The load structure follows industrial EAF operating practices characterized by short-duration but electricity-intensive melting campaigns and continuous auxiliary consumption. These values are consistent with ranges reported by (World Steel Association 2021), (Zhuo, Harbi and Pistorius 2025) (Joint Research Center (JRC) 2012) studies, which identify EAF steelmaking as highly electricity-intensive but suitable for industrial demand response and renewable-integration applications. The calibration and the decomposition of the electrical loads are presented in Table 2.

4.5 Thermal Loads Calibration

The thermal-demand coefficients of the modeled Electric Arc Furnace (EAF) steel-production route were calibrated using a hybrid engineering–literature methodology combining reheating-furnace studies, hot-rolling analyses, industrial energy balances, and process-integration research. Downstream rolling, reheating, and finishing stages were parameterized using literature-reported thermal intensities of approximately 1.0–1.8 GJ/t steel, with representative values near 1.5–1.6 GJ/t based on the (World Steel Association 2021), (Oxmaint 2026), (Mc Brien, Serrenho and Allwood 2016) studies, the (Schneider and Lechtebohmer 2018) reports, and reheating-furnace integration literature study (Rosado, et al. 2020). Lower-load operational states were modeled using simplified fixed thermal loads representing compressors, pumps,

ventilation, standby burners, and auxiliary industrial services. EAF melting stages were intentionally excluded from the thermal-buffer balance because EAF steelmaking is fundamentally modeled as an electricity-driven metallurgical process, consistent with (World Steel Association 2021) studies reporting electricity intensities of approximately 350–450 kWh/t steel for scrap-based EAF production routes. The calibration of the thermal loads is presented in Table 2.

Table 2. Hourly Electrical Load Decomposition, Operational Sequencing, and Thermal Demand Profile of the EAF Mini-Mill Production Process

Process Hour,t	Critical MW	EAF MW	Secondary/Rolling MW	Total MW	Operational Support	Heat Required (MWh)
0	12	0	0	12	idle auxiliaries only	1,03
1	12	0	0	12	Idle	0,62
2	12	0	0	12	Idle	0,62
3	12	0	0	12	Idle	0,62
4	12	0	0	12	minimum plant load	2,89
5	12	0	3	15	preparation starts	2,06
6	12	0	12	24	compressors, scrap handling, pre-start	0,52
7	12	0	26	38	warm-up and line preparation	4,13
8	12	60	0	72	EAF heat starts	
9	12	60	6	78	EAF + secondary support	0,41
10	12	60	8	80	peak melting/refining	
11	12	60	4	76	EAF taper + downstream overlap	0,83
12	12	0	16	28	casting / rolling / auxiliaries	32,18
13	12	0	12	24	downstream processing continues	30,94
14	12	0	8	20	reduced downstream load	15,47
15	12	60	0	72	second EAF campaign starts	
16	12	60	6	78	active melting	
17	12	60	8	80	peak second campaign	
18	12	60	0	72	still on, lower overlap	0,83
19	12	0	20	32	downstream finishing	7,22
20	12	0	14	26	rolling / handling	30,94
21	12	0	8	20	late finishing	22,69
22	12	0	3	15	shutdown sequence	1,34
23	12	0	0	12	idle auxiliaries	1,13

4.6 Scrap Rate calculation interpretation

The coefficients of the scrap function were calibrated using ranges and qualitative relationships reported in steel manufacturing literature. Baseline scrap rates were set within the 1–3% range observed in EAF-based production (World Steel Association 2021); (Joint Research Center (JRC) 2012). Load ramping and instability effects were incorporated based on operational constraints highlighted in industrial demand response studies, which emphasize on the operational inefficiencies and quality risks, due to the fact that steel plants cannot arbitrarily change load thanks to process constraints and the necessity for a metallurgical stability (Castro, Sun and Harjunkoski 2013) (Zhang, et al. 2015)). In addition, in (Zhang, et al. 2015) within the constraints discussion in the problem formulation section, ramp penalty was supported by the fact that the industrial processes have limited flexibility due to operational constraints. Therefore, if the load has a sudden change of 20 MW, then $\Delta scrap,1 = 0,00015 * 20 (MW) = 0,003 = 0,3\%$

For EAF switching (start/stop) effects there was the (Joint Research Center (JRC) 2012), in the section of EAF operation cycle, which emphasizes limited flexibility and sensitivity to abrupt changes. According to this study, since EAF operates in discrete heat cycles, the consequent

interruptions affect the thermal balance and the energy efficiency. In addition, in (International Energy Agency-IEA 2020), within the section of the EAF technology characteristics, the EAF operation is batch-based and thermally intensive repeated start-stop cycles reduce efficiency. This also supports the switching penalty magnitude. For thermal socks and restart losses, $0,0009 * \{EAF,t - EAF, t - 1\}$, where EAF,t and $EAF, t - 1$ are binary switching, either with $1 \rightarrow 0$ or with $0 \rightarrow 1$, then $\Delta scrap,2 = 0,0009 = 0,09\%$

Off-design operation was assigned a smaller coefficient reflecting its secondary impact on process efficiency in: a) (International Energy Agency-IEA 2020) within the section of Process efficiency and optimization, claiming that the optimal operation occurs near design conditions and that the part-load operation reduces efficiency; b) (Bruckner, et al. 2015) within the section of industrial process integration, pointing out that the non-optimal operation reduces the efficiency and the recovery potential. Then for deviation of nominal operation, i.e. if the plant runs at 80% of the. Maximum load, then $1 - 0,8 = Load\% - 1$, then $Load\% - 1 = 0,2$ and $\Delta scrap,3 = 0,0004 * 0,2 = 0,0008\%$

Temperature deviations were identified as the primary driver of scrap generation, consistent with process control requirements in electric arc furnaces and reheating systems (European Commission, 2012). This study claimed that is very important to maintain the proper temperature, which is critical to product. The quality deviations lead to off-spec steel, to reprocessing and finally to the increased scrap. In addition, in (Johanson and Soderstrom 2011), within the energy efficiency section, the thermal coefficient dominance is supported by the consequences of the temperature instability, such as the quality loss and the increased material losses. Assuming the thermal deficit to be at 10 MW, then $\Delta scrap,4 = 0,0003 * 10 = 0,3\%$.

The resulting parameterization ensures realistic magnitudes and relative importance of scrap drivers. Table 3 maps the coefficients of scrap rate to each driver.

Table 3. Mapping of scrap rate coefficients to each driver

Mapping of scrap rate coefficients to each driver	
Coefficient	Value
Baseline	0.021
Load ramp	0.00015
EAF switching	0.0009
Off-design	0.0004
Thermal deficit	0.0003

$$scrap, rate = 0,021 + 0,00015 * \Delta Load + 0,0009 * \{EAF,t - EAF, t - 1\} + 0,0004 * \{1 - Load\%\} + 0,0003 * \{\max(0, Q_{required} - Q_{thermal}^{available})\} \quad (1)$$

4.7 Solar irradiation (March, Athens) and PV Generation

The dataset concerning Solar Radiation availability by region, month, and day is obtained from (Koudouris, et al. 2017). This data set is presented in the data link mentioned in the Dataset Availability Statement. The city of Athens is selected as the reference region.

4.8 Electricity and Natural Gas prices

Electricity price data for 2025 for Greece are downloaded from (ENTSO-e 2025). The Natural Gas price dataset is obtained from the (Exchange 2025). Both sets, as cleansed data after the extraction from the above-mentioned resources are presented in the data link mentioned in the Dataset Availability Statement

4.9 The integration of electricity and fuels markets dynamics and the variables contribution of the renewables plus the energy recovery

For each actual chronological market hour $\tau_{d,t}$ the optimization procedure recalls every hourly electricity or natural gas price and solar availability. Therefore, the production costs are evaluated by the utilization of the actual hourly market conditions and the optimization captures the intraday volatility of electricity price and the day to day natural gas price variations, interacting with the corresponding electricity price. The intermittent renewable generation is also depicted and the intraday temporal arbitrage opportunities, as well. That is very important for the industrial demand response function, in which the process stage shift can reflect the low prices of electricity during the night, the high penetration of renewable generation during specific time windows and the natural gas price intervals.

Hourly photovoltaic generation is calculated endogenously using hourly solar availability values. PV generation equation (KWh) is given by:

$$PV_{d,t} = \frac{S_{\tau_{d,t}} * n_{PV,eff} * A_{PV}}{1000} \quad (6)$$

Where: $S_{\tau_{d,t}}$ represents the solar availability in KW/m², $n_{PV,eff}$ is the efficiency of a PV panel, equal to 18%, A_{PV} is the installation area for PVs in m², which is equal to 66.700 m².

The net grid electricity demand is given by:

$$GI_{d,t} = L_{d,t} - PV_{d,t} - B_{d,t}^{dis} \quad (7)$$

Where, $GI_{d,t}$ reflects the overall Grid Import or else, the Net Grid electricity demand after considering the electricity demand for steel production $L_{d,t}$, the PV generation $PV_{d,t}$, and the electricity that is supplied through the battery discharge $B_{d,t}^{dis}$, that is utilized from the storage system installed. It is easily concluded that higher solar availability directly lowers electricity purchasing cost.

The battery storage system is modeled explicitly in order to exploit temporal electricity price arbitrage. The battery can charge during low-price periods, or discharge during high-price periods, therefore it can reduce exposure to expensive electricity market hours.

The battery state of charge evolves according to:

$$SOC_t = SOC_{t-1} + n_b * B_t^{ch} - \frac{B_t^{dis}}{n_b} \quad (8)$$

Where n_b is the battery efficiency. To ensure cyclic operation of each 24-hour production circle, the following constraint is imposed:

$$SOC_{t=1} = SOC_{t=24} \quad (9),$$

meaning that each circle begins and ends with identical storage conditions. The battery therefore acts as a flexibility mechanism capable of shifting electricity consumption away from expensive market periods.

Recovered industrial heat from EAF and downstream stages is integrated into a thermal buffer system. Recovered heat is stored and later utilized to offset thermal energy requirements in subsequent production stages. The thermal buffer state evolves according to:

$$SOC_{th,t} = SOC_{th,t-1} + R_t^{heat} - T_t^{dis} \quad (10)$$

where:

$SOC_{th,t}$ and $SOC_{th,t-1}$ are thermal buffer state, in production time t and $t-1$

R_t^{heat} is the generation from recovered heat

T_t^{dis} is the thermal buffer discharge

The thermal discharge reduces natural gas consumption required for industrial heating processes. The effective thermal demand covered by natural gas becomes:

$$H_{d,t}^{gas} = H_{d,t} - T_{d,t}^{dis} \quad (11)$$

Where, $H_{d,t}$ is the total thermal requirement. Consequently, the thermal buffer acts as an industrial heat recovery mechanism that lowers fossil fuel consumption and operating costs.

4.10 Operational interpretation, penalties and scrap rates

The model also integrates scrap-rate related operational penalties. Industrial process rescheduling may increase operational stress, thermal instability, or quality deviations if production stages are excessively shifted or interrupted.

The economic impact of scrap generation is represented through the scrap-related cost component:

$$C_{d,t}^{scrap} = scrap_rate_{d,t} * c^{scrap} * Q_{d,t} \quad (14)$$

Where, $scrap_rate_{d,t}$ is the scrap generation rate, denoting the scrap fraction generated at process stage t during process circle d , c^{scrap} is the unit scrap cost and $Q_{d,t}$ is production quantity associated with the corresponding process stage. This term prevents unrealistic over-optimization of production schedules by penalizing excessive operational deviations. Therefore, the optimization balances: a) electricity cost reduction, b) renewable energy utilization, c) storage flexibility, d) thermal recovery, and e) production quality preservation. The Scrap Rate is therefore integrated directly into the optimization objective function as an operational penalty term.

4.11 Optimization Objective and techno-Economic Interpretation of the Objective Function

The optimization problem is formulated as a start-time scheduling problem over multiple operational cycles. The framework includes the **Decision Variable** is the selected process start hour h_d , for each selected calendar day d . The optimization horizon is upon 8 selected calendar days plus the additional spill-over day, including a chronological evaluation. The process constraints determine that each process circle lasts exactly 24 consecutive process hours, follows a strict chronological continuity and does not wrap within the same calendar day in any circular mode.

The **objective function** is based on the optimization framework which balances: a) electricity cost minimization, b) renewable energy utilization, storage flexibility, thermal recovery, and industrial product quality preservation.

A generalized total operating cost formulation may be written as:

$$\min \sum_{d=1}^8 \sum_{t=1}^{24} \{ (L_{d,t} - PV_{d,t} - B_{d,t}^{dis} + B_{d,t}^{ch}) * p_{\tau,d,t}^{el} + (H_{d,t} - T_{d,t}^{dis}) * p_{\tau,d,t}^{gas} + C_{d,t}^{scrap} \} \quad (15)$$

The optimization minimizes total operational cost while maximizing industrial flexibility through photovoltaic self-consumption, battery arbitrage, recovered-heat utilization, and dynamic process scheduling. The framework integrates electricity-market interaction, renewable integration, thermal recovery, and industrial continuity within a unified techno-economic optimization environment. Chronological scheduling aligns electricity-intensive stages with favorable market conditions while preserving metallurgical stability, operational feasibility, and industrial process continuity.

4.2.6 Operational Constraints

Binary decision variables are introduced for battery charging/discharging, thermal discharging.

Operational constraints include:

$$\sum \text{Battery Charge}_t \geq 4 \quad (16)$$

$$\sum \text{Thermal Discharge}_t \leq 3 \quad (17)$$

$$\sum \text{Thermal Discharge}_t < 9 \quad (18)$$

for each 24-hour process circle. These constraints preserve technical realism and prevent excessive switching behavior.

The framework minimizes total industrial operating cost through integrated optimization of electricity procurement, photovoltaic self-consumption, battery-storage operation, thermal-buffer utilization, and chronological process scheduling. Electricity-intensive EAF stages are dynamically aligned with favorable market conditions while preserving metallurgical stability and industrial continuity. The optimization incorporates three major operational constraints: battery charging/discharging limits (16), thermal-buffer discharge limits (17), and process-sequence continuity constraints (18). Consequently, the framework captures renewable integration, industrial demand response, waste-heat recovery, operational circularity, and process-quality preservation within a unified techno-economic MILP environment suitable for low-carbon EAF-based steel production systems operating under volatile electricity-market conditions.

4.12 Operational Circularity Index

To evaluate the operational circularity performance of the proposed industrial prosumer framework, a set of integrated circularity and environmental indicators is introduced. Unlike conventional circular economy metrics that focus primarily on material recycling flows, the proposed methodology extends circularity assessment to the operational and energy-management level. The selected indicators, as presented in Table 4, quantify the degree to which renewable electricity, recovered process heat, flexible production scheduling, and material efficiency contribute to reducing external energy dependence, minimizing waste generation, and improving process stability. In particular, the framework evaluates renewable electricity utilization through on-site photovoltaic self-consumption, internal thermal energy recirculation through waste-heat recovery, emissions intensity per tonne of steel product, and material efficiency through scrap minimization. Together, these indicators form a multidimensional representation of “operational circularity,” linking industrial demand response, energy flexibility, thermal integration, and production-quality preservation within a unified optimization framework.

Table 4. Operation Circularity Index, with the Interpretation and the CE dimension

Indicator	Symbols	Unit	Interpretation	Circular Economy Dimension
Renewable Electricity Share	RESshare	%	Share of plant electricity demand covered by on-site photovoltaic generation plus the discharged energy from the battery storage took place in earlier time slots	Circular energy input
Recovered Heat Utilization Rate	RHU	%	Fraction of waste heat recovered from metallurgical processes that is reused in reheating or thermal processes	Internal energy circularity (how much waste heat is reused internally)

Electricity Emissions Intensity	EI	tCO ₂ /t steel	Emissions associated with electricity consumption plus natural gas consumption, per tonne of steel product	Environmental efficiency
Material Efficiency	ME	%	Share of input steel material converted into final product	Material circularity
Scrap Generation Rate	Scrap,rate	%	Steel losses during processing caused by operational disruptions or temperature deviations	Production stability / material efficiency
Circularity Performance Index	CPI	dimensionless	Composite index combining renewable energy use, heat recovery, and scrap minimization	Overall operational circularity

$$RES_{share} = \frac{\sum_t (PV_t + B_t^{dis})}{\sum_t D_t} * 100 \quad (19)$$

$$RHU = \frac{\sum_t Q_t^{reused}}{\sum_t Q_t^{recovered}} * 100 \quad (20)$$

$$EI = \frac{\sum_t ((D_{grid} * EF_{1,t}) + (NG_t * EF_{2,t}))}{Steel\ Production\ Capacity} \quad (21)$$

$$ME = 1 - (scrap,rate) \quad (22)$$

$$scrap,rate = \frac{\sum_t scrap_t}{\sum_t Production_t} \quad (23)$$

$$CPI = w_1 * RES_{share} + w_2 * RHU + w_3 * (1 - scrap,rate) \quad (24)$$

In the base case, equal or near-equal weights are adopted to avoid imposing strong subjective preferences among the three circularity dimensions, as shown in Table 5. A slightly higher weight is assigned to heat recovery and scrap minimization because these indicators represent internal circularity mechanisms directly linked to resource efficiency and material loss reduction. Sensitivity analysis is then performed by varying the weights to test the robustness of the Circular Performance Index (CPI).

Table 5. weight factors to incorporate a) renewable, b) circularity, c) material priorities within the Circular Performance Index (CPI) calculation

Case	(w_1)	(w_2)	(w_3)
Equal weights	0.333	0.333	0.333
Renewable-priority	0.50	0.25	0.25
Circularity-priority	0.25	0.375	0.375
Material-priority	0.20	0.30	0.50

4.13 Economic Indicator Index

To complement the operational circularity assessment, the proposed framework introduces a set of economic indicators that quantify the contribution of renewable integration, storage technologies, thermal energy recovery, and market-oriented operation to industrial cost

minimization. These indicators, as shown in Table 6, evaluate the relative contribution of each flexibility mechanism to the optimized operational cost of the steel plant. In particular, the methodology measures the share of total cost reduction attributable to on-site renewable electricity generation, battery storage operation, and thermal buffering through waste-heat recovery. Renewable electricity contribution reflects the fraction of electricity expenditure avoided through photovoltaic self-consumption and renewable-based battery discharge, while the battery contribution captures the economic value of temporal electricity shifting and peak-load management. Similarly, the thermal-buffer contribution quantifies the reduction in electricity or fuel costs achieved through internal heat reuse and thermal storage utilization. Together, these indicators provide a multidimensional economic evaluation of industrial prosumer operation, linking demand response, energy storage, renewable integration, and thermal circularity with operational profitability and electricity market participation.

Table 6. Economic Indicator Index and its interpretation

Indicator	Symbol	Unit	Interpretation
Contribution of Renewables to Total Cost Minimization	$RES_{c,min,s}$	%	Share of cost reduction due to renewables (PV + battery use) contribution compared to the total (minimized) cost
Contribution of Battery use to Total Cost Minimization	$B_{c,min,s}$	%	Share of cost reduction due to the battery use contribution compared to the total (minimized) cost
Contribution of Heat re-use to Total Cost Minimization	$Q_{c,min,s}^{reused}$	%	Share of cost reduction due to the heat re-use contribution compared to the total (minimized) cost
Utility function of steel Producer	UF	€/MWh	utility function of steel manufacturer calculating revenues per MWh consumed minus the total minimized cost per MWh consumed

$$RES_{c,min,s} = \frac{\sum_t \{(D_t - D_{grid,t}) * Pr_{el,t}\}}{\sum_t C_t} * 100 \quad (24)$$

$$B_{c,min,s} = \frac{\sum_t \{(D_t - D_{grid,t} - PV_t) * Pr_{el,t}\}}{\sum_t C_t} * 100 \quad (25)$$

$$Q_{c,min,s}^{reused} = \frac{\sum_t (Q_{t}^{reused}) * Pr_{NG,t}}{\sum_t C_t} * 100 \quad (26)$$

In addition, the framework incorporates a utility function of the steel producer that represents the economic benefit derived from optimized production scheduling under dynamic electricity prices, renewable availability, and operational flexibility constraints.

The producer utility function is formulated as a net economic utility per unit of required industrial energy use. The first term represents the production value of useful steel output, adjusted for scrap generation, and normalized by the technologically required process load. This term is independent of any excess electricity demand because steel output is determined

by the required load of each production stage rather than by additional electricity consumption. The second term represents the operating cost per unit of grid electricity demand. Total electricity demand is defined as:

$$D_{grid,t} = L_t^{req} + x_t \quad (27)$$

where L_t^{req} is the required process load and x_t denotes any additional excess electricity use that does not contribute to production. Since x_t increases cost without increasing useful output, the economically rational solution is $x_t^*=0$, while flexibility is expressed through the temporal scheduling of L_t^{req} not through increasing electricity consumption.

This formulation reflects the inelastic nature of industrial electricity demand: the plant does not demand more electricity because prices are low, nor can it freely reduce required process loads when prices are high. Instead, the optimization problem concerns the timing of technologically required loads across market hours.

$$UF = Revenues - Total Cost = \frac{\sum_t Prod * (1 - scrap, rate) * Pr_{steel}}{\sum_t L_t^{req}} - \frac{\sum_t C, min, t}{\sum_t (D_{grid})} \quad (28)$$

The (industrial consumer buys up to the point that the Marginal Utility function of the good (electricity) equals to the Price, therefore:

$$\frac{\partial U}{\partial D_{GRID}} = Price, el \quad (29)$$

$$\frac{\partial}{\partial D_{GRID}} \left\{ \frac{\sum_t Prod * (1 - scrap, rate) * Pr_{steel}}{\sum_t L_t^{req}} - \frac{\sum_t C, min, t}{\sum_t (D_{grid})} \right\} = Price, el \quad (30)$$

The first utility component is independent of the demand decision variable because steel output is constrained by the required technological load of the production process. Therefore, additional electricity demand does not generate additional productive output, implying that the derivative of the first utility term with respect to demand is equal to zero (0). Therefore:

$$D_{grid}(Pr, ele) = \sqrt{\frac{\sum_t C, min, t}{Price_{el}}} \quad (31)$$

The electricity-demand representation adopted in the present framework is defined at the level of the 24-hour industrial production cycle rather than over the aggregated 168-hour weekly horizon. This modelling choice reflects the technological structure of steel manufacturing systems, where production loops, required process loads, EAF melting campaigns, downstream processing sequences, scrap-related penalties, and operational continuity constraints are inherently organized within repeated daily production cycles.

The utility function is defined at the level of each 24-hour industrial production circle, where production output, process loads, scrap penalties, and energy costs are technologically linked. The weekly utility is then obtained by aggregating the utility values across the seven consecutive production circles. This formulation preserves the economic interpretation of producer utility while recognizing that industrial electricity demand is largely inelastic in quantity and flexible mainly in its temporal allocation.

The aggregate utility over the 168-hour weekly horizon is calculated using the cumulative sums of the economic and energy-related parameters across the full optimization period rather than as a simple arithmetic sum of the individual 24-hour utility values. This approach provides a weighted representation of weekly producer utility per unit of energy use, capturing the complete weekly production, cost, and electricity-consumption balance within the industrial system. Consequently, the weekly utility indicator is interpreted as an aggregate techno-economic performance metric reflecting the overall operational efficiency of the industrial prosumer system under dynamic electricity-market conditions, rather than as a direct accumulation of daily utility indices.

4.14 Scenarios Descriptions

Three representative electricity-market scenarios were evaluated in order to investigate the operational behavior of the industrial prosumer model under contrasting market conditions: Low Electricity Price Scenario (27 April–3 May 2025), High Electricity Price Scenario (11–18 February 2025), and Random Period Scenario (19–25 March 2025).

4.15 Electricity and Natural Gas Emission Factors in Greece (EF1 & EF2)

The EF1 factor for electricity consumption in Greece for 2024 is calculated using the official methodology of the (Energy 2025) combined with (DAPEEP 2024) residual energy-mix data, Guarantees of Origin (GOs), and European Attribute Mix information. For EF2, related to natural gas combustion in boilers and stationary thermal systems, the methodology follows IPCC and national emission-factor guidance using the Net Calorific Value (NCV) of natural gas together with fuel-specific CO₂, CH₄, and N₂O emission factors. Main data sources include the (Energy 2025) and (DAPEEP 2024) residual-mix reports.

5. Results

5.1 Economic Indicators

5.1.1 Renewables contribution to cost minimization.

The contribution of renewables, defined as the combined effect of PV generation and PV-based battery use, differs substantially across the three scenarios. It is highest in the **Low Electricity Price Scenario** at **7.20%**, remains relatively strong in the **Random Period Scenario** at **6.79%**, and falls to **4.27%** in the **High Electricity Price Scenario**. This suggests that renewable integration contributes more effectively when electricity-market conditions allow better temporal alignment between PV availability, production scheduling, and storage use.

5.1.2 Battery contribution alone.

The standalone battery contribution is more limited than the total renewable contribution. It reaches **2.75%** in the low-price scenario, declines to **1.49%** in the random scenario, and becomes very small in the high-price scenario at **0.59%**. This indicates that, under the PV-only charging constraint, battery value depends strongly on the availability of surplus solar generation and on whether stored PV electricity can be discharged during economically relevant hours.

5.1.3 Heat reuse contribution.

The contribution of heat reuse to cost minimization is highest in the **Low Electricity Price Scenario** at **2.86%**, moderate in the **High Electricity Price Scenario** at **1.30%**, and almost negligible in the **Random Period Scenario** at **0.04%**. This suggests that thermal buffering creates value only when recovered heat is sufficiently synchronized with useful thermal

demand, such as reheating, finishing, drying, or process-temperature support. The results are presented in Figure 1.

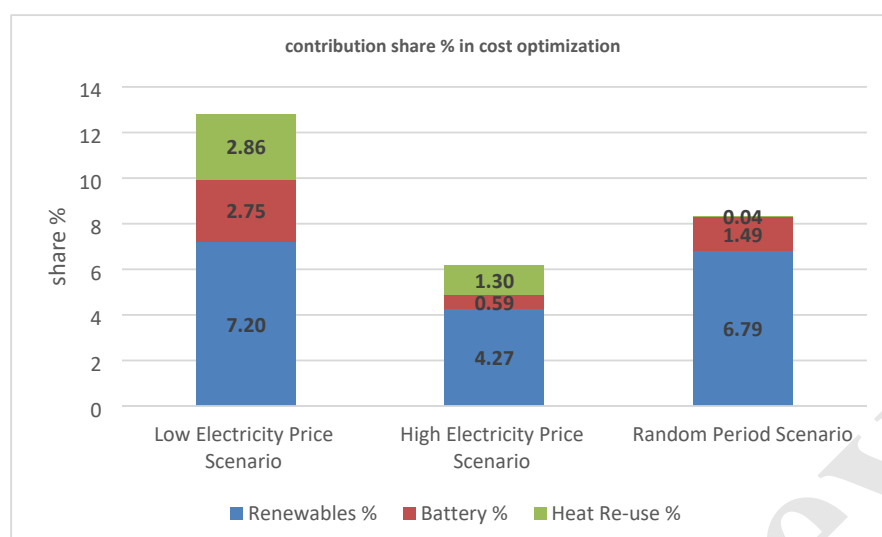


Figure 1. Contribution of a) renewables, b) battery standalone and c) heat re-use in the cost optimization, across 3 scenarios regarding different electricity prices time slots

5.1.4 Utility function and UF components.

The producer utility function is highest in the **Low Electricity Price Scenario** at **1173.72 €/MWh**, followed by the **Random Period Scenario** at **1020.97 €/MWh**, and lowest in the **High Electricity Price Scenario** at **941.84 €/MWh**. This indicates that the producer obtains the greatest net operational benefit when electricity prices are low and flexible scheduling can reduce procurement costs. Conceptually, **UF1**, representing revenues or production value, remains linked to the useful output of the plant, while **UF2**, representing costs, increases under high-price conditions. Therefore, the decline of total UF in the high-price scenario is mainly driven by the higher cost component rather than by a reduction in production value. The UF decomposition is shown in Figure 2.

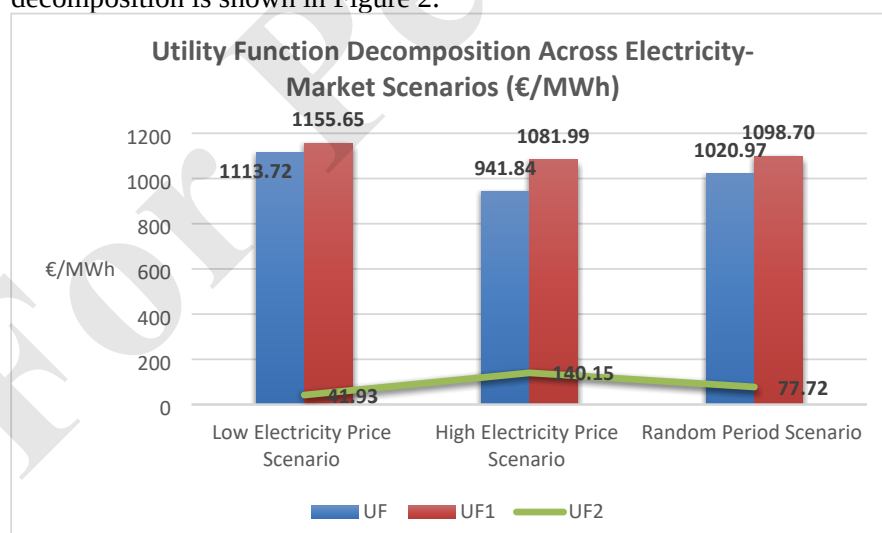


Figure 2. Utility Function Decomposition Across Electricity-Market Scenarios

5.1.5 Total 7-day cost across scenarios.

The total minimized cost over the full 7-day period is lowest in the **Low Electricity Price Scenario**, equal to **€287,254.83**. The **High Electricity Price Scenario** rises sharply to **€1,025,365**, while the **Random Period Scenario** is somewhere in the middle at **€559,409**. This confirms that electricity-price selection strongly affects total industrial operating cost, even when PV, battery storage, heat reuse, and scrap-related penalties are included in the optimization. The above figures are presented in Figure 3.

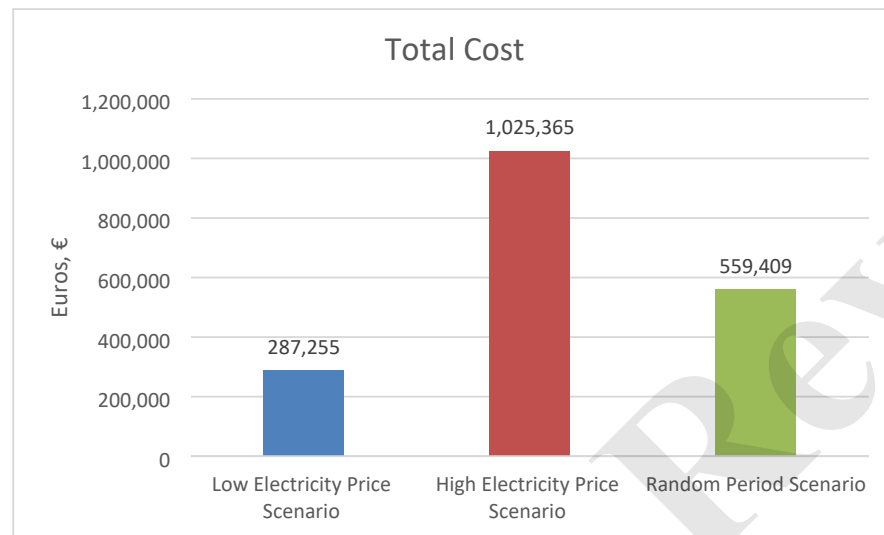


Figure 3. Total cost across 3 different electricity – market scenarios

5.1.6 Daily cost evolution across the 7 days.

In the low-price scenario, daily costs remain relatively moderate, ranging from **€21,241** to **€53,261**, with the lowest cost on day 5 and the highest on day 2. In the high-price scenario, daily costs are consistently elevated, ranging from **€120,253** to **€175,714**, showing persistent market pressure throughout the selected week. The random scenario presents a similar range like the high-price scenario: from **€52,972** and **€118,766** thereafter. **This indicates that even random periods may contain isolated high-cost or low-cost days, by coincidence, however they follow the trend, depending on how closely to the high or low-cost case they run.** These are presented in Figure 4.

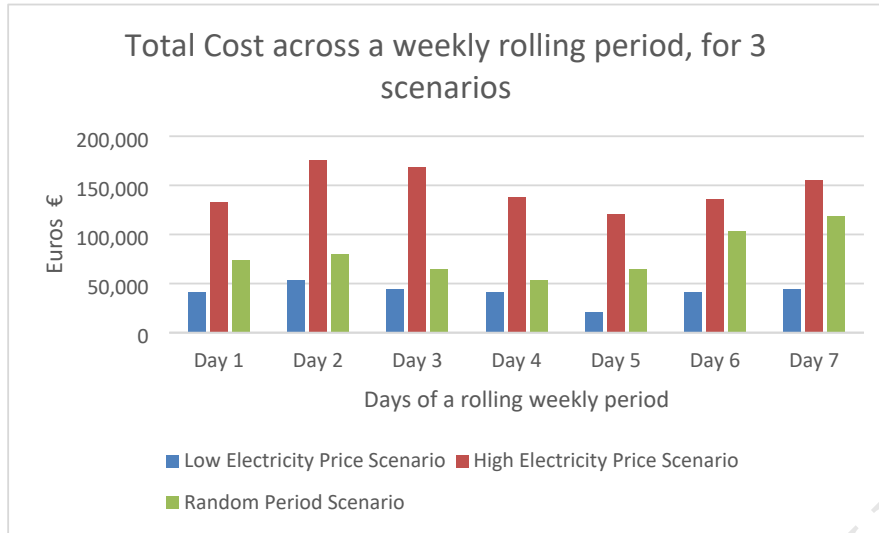


Figure 4. Total Cost across a weekly rolling period across 3 Electricity - market scenarios

5.1.6 Demand Function

The demand-function results demonstrate a strong inverse relationship between electricity price and industrial electricity demand across all examined scenarios, consistent with the theoretical structure of the proposed utility framework. In all three electricity-market scenarios, the demand curves exhibit a nonlinear downward-sloping behavior, where electricity demand decreases as electricity prices increase. However, unlike classical residential demand functions, the observed response is driven primarily by operational scheduling optimization and temporal allocation of industrial process loads rather than by unrestricted consumption reduction.

The Low Electricity Price Scenario, as presented in Figure 5, exhibits the steepest curvature and the highest sensitivity at low-price regions, where demand rises sharply as prices approach very low levels, reflecting the economically optimal concentration of electricity-intensive production stages during favorable market periods.

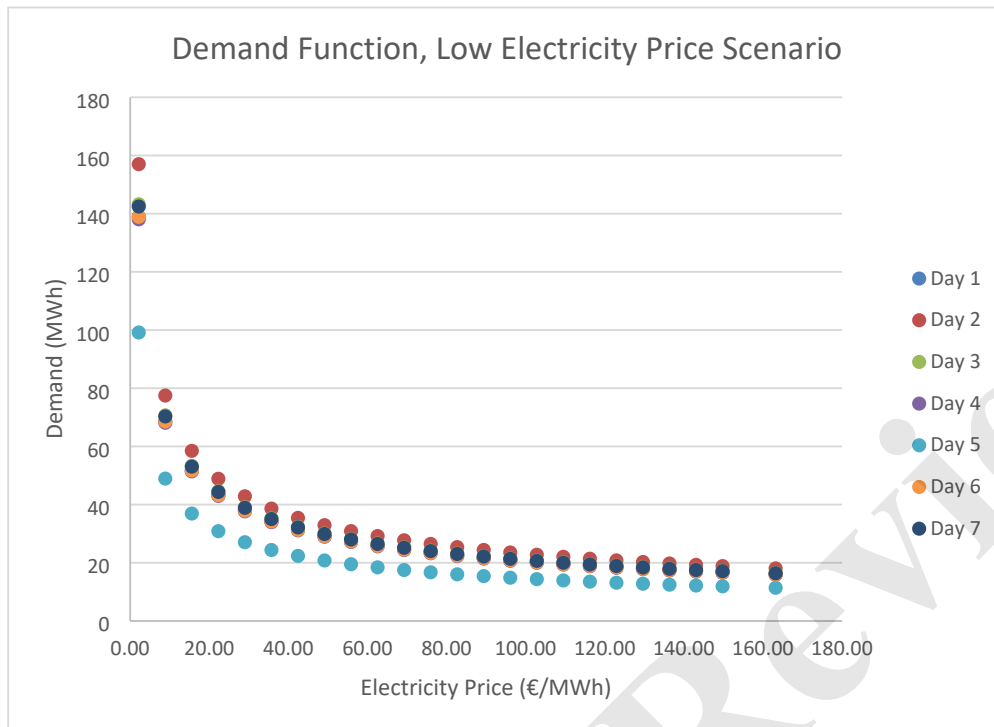


Figure 5. Demand curve with regards to Low Electricity Price Scenario

In contrast, the High Electricity Price Scenario, as shown in Figure 6, presents flatter demand trajectories with lower absolute demand levels, indicating constrained industrial flexibility under persistently expensive electricity conditions.

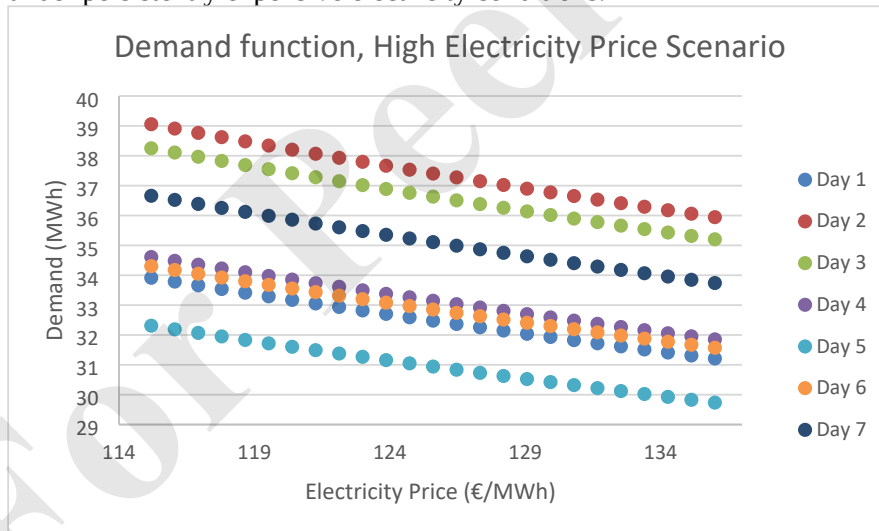


Figure 6. Demand curve with regards to High Electricity Price Scenario

The Random Period Scenario, as presented in Figure 7, exhibits intermediate behavior with moderate variability between daily demand curves.

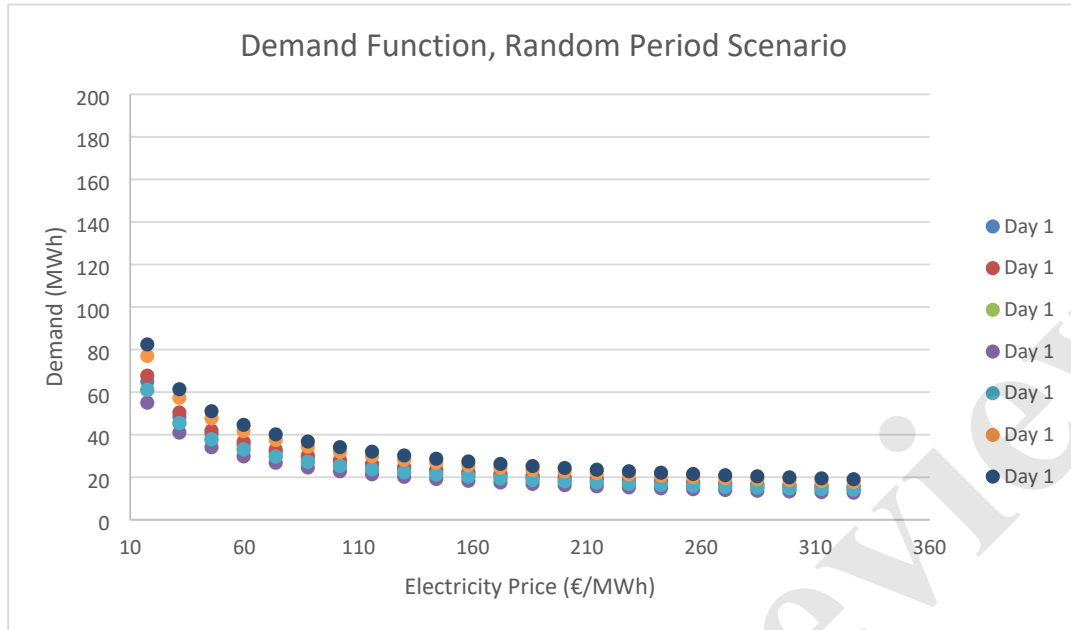


Figure 7. Demand curve with regards to Random period scenario

Across all scenarios, the curves gradually flatten as electricity prices increase, revealing a decreasing marginal response of demand to further price increases. Importantly, demand does not converge fully to zero even under very high electricity prices, reflecting the technologically required and largely inelastic nature of industrial process loads within EAF-based steel production systems. Furthermore, daily shifts among the demand curves indicate that variations in operational costs, renewable availability, and process conditions influence the vertical positioning of the demand function throughout the weekly optimization horizon.

5.2 Environmental Indicators

5.2.1 Renewable Electricity Share.

The Renewable Electricity Share exhibits significant variation across the three electricity-market scenarios, reflecting the changing ability of the industrial prosumer system to utilize locally generated renewable electricity. As shown in Figure 8, the highest renewable share is observed in the Low Electricity Price Scenario, reaching **8.89%**, followed by the Random Period Scenario at approximately **6.74%**, while the High Electricity Price Scenario records the lowest renewable share at **4.72%**. The higher renewable penetration during the low-price period indicates improved synchronization between photovoltaic generation, battery charging/discharging behavior, and flexible industrial scheduling. In contrast, during the high-price scenario, operational constraints and reduced alignment between solar availability and production demand limit the contribution of renewable electricity to total plant consumption. These results demonstrate that industrial flexibility and storage operation are critical for maximizing renewable self-consumption within steel-sector prosumer systems.

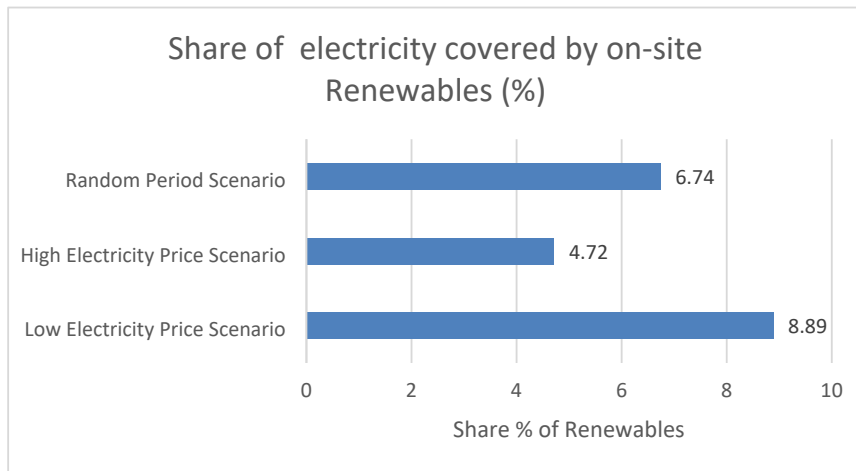


Figure 8. Share (%) of electricity covered by on -site renewables, across Electricity - market scenarios

5.2.2 Recovered Heat Utilization (RHU).

The Recovered Heat Utilization (RHU) indicator remains consistently high across all scenarios, highlighting the important role of thermal buffering and waste-heat reuse within the proposed operational circularity framework. As shown in Figure 9, the highest RHU value is observed in the High Electricity Price Scenario at 37.47%, followed closely by the Low Electricity Price Scenario at 35.84% and the Random Period Scenario at 35.63%. The slightly higher RHU during periods of elevated electricity prices suggests that the optimization algorithm intensifies internal heat reuse in order to reduce dependence on costly electricity or fuel-based reheating. This confirms that thermal energy recovery becomes economically more valuable during stressed electricity-market conditions. Overall, the results demonstrate that thermal buffering significantly enhances internal energy circularity by enabling recovered metallurgical heat to substitute part of the downstream thermal demand.

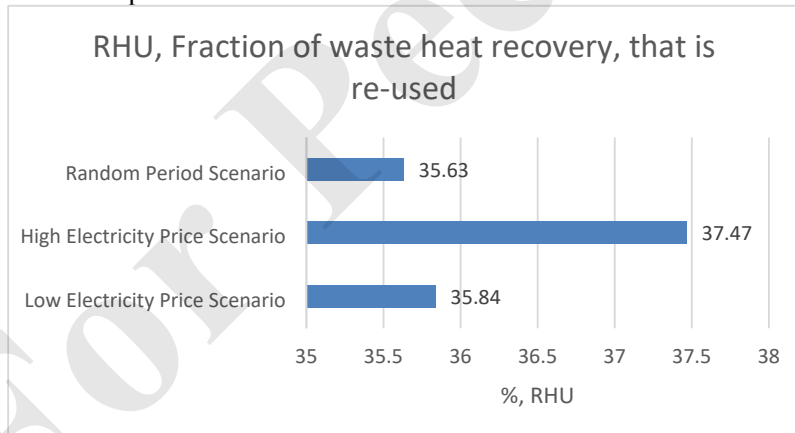


Figure 9. Fraction of waste heat recovery that is re-used across Electricity - market scenarios

5.2.3 Energy Emissions Intensity (EI).

The Energy Emissions Intensity (EI), expressed in Kg CO₂ /t of steel product, reveals important differences in environmental performance across the three scenarios. **They are associated with the electricity plus natural gas consumption.** As presented in Figure 10, the lowest emissions intensity is achieved in the Random Period Scenario at approximately **141.11** Kg CO₂ /t of steel, indicating more favorable alignment between electricity consumption and lower-carbon

electricity periods. The Low Electricity Price Scenario follows with **158.28** Kg CO₂ /t of steel, while the High Electricity Price Scenario exhibits the highest emissions intensity at approximately **177.83** Kg CO₂ /t of steel. These findings indicate that periods characterized by high electricity prices are also associated with more carbon-intensive electricity generation, likely due to increased dependence on natural gas or other fossil-based marginal generation technologies. Consequently, flexible scheduling, renewable integration, and thermal buffering not only reduce operational costs but also improve environmental performance by shifting industrial demand away from carbon-intensive electricity periods.

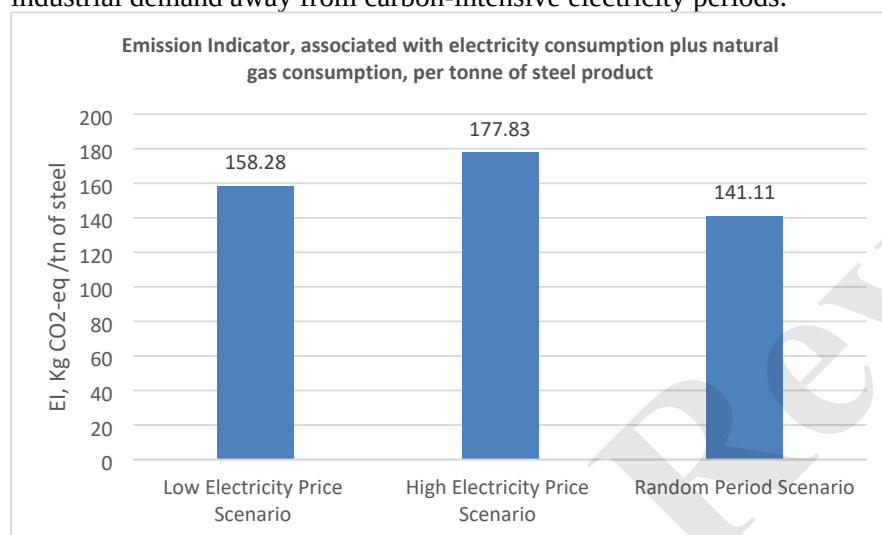


Figure 10. Emission Indicator, associated with electricity consumption plus natural gas consumption, per tone of steel product.

5.2.4 Scrap Generation Rate.

The Scrap Generation Rate remains relatively stable across the three scenarios, demonstrating that the proposed demand-response and operational-flexibility strategies do not significantly compromise process quality or production stability. As shown in Figure 11, the lowest scrap rate is observed in the High Electricity Price Scenario at **2.30%**, followed closely by the Low Electricity Price Scenario at **2.40%** and the Random Period Scenario at approximately **2.41%**. The small variation between scenarios indicates that the optimization framework successfully preserves metallurgical and operational constraints while implementing flexible scheduling strategies. This is particularly important because excessive load shifting or thermal instability could otherwise increase material losses and undermine operational circularity objectives. The results therefore confirm the effectiveness of the “quality-preserving curtailment” concept introduced in the study, whereby demand flexibility is implemented without generating excessive scrap or process disruptions.

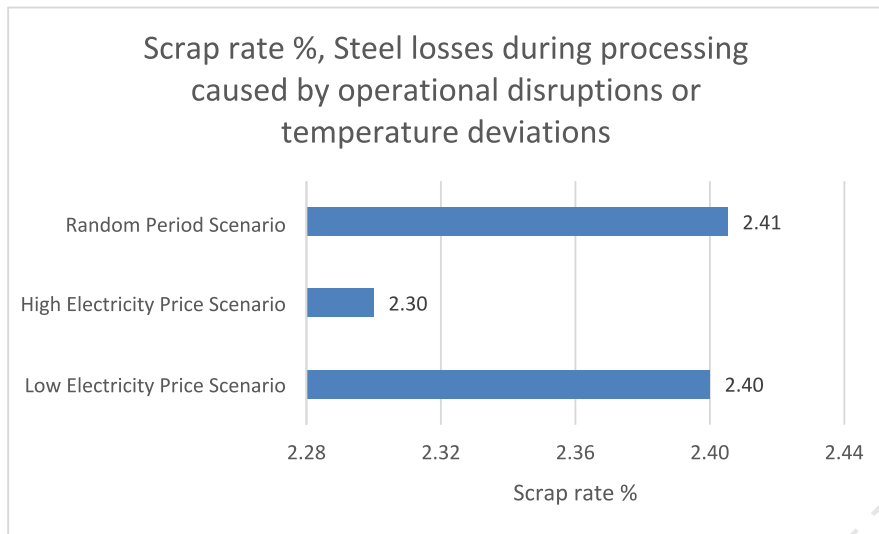


Figure 11. Scrap rate (%) across Electricity - market scenarios

5.2.5 CPI across weights and scenarios.

The Circularity Performance Index changes depending on the weighting scheme w1, w2, w3 . Under **equal weights**, CPI is highest in the low-price scenario (**14.43**), lowest in the high-price (**13.62**) and slightly higher in the random (**13.64**) scenario. The highest overall CPI appears under the **Circularity-priority** weighting, reaching **15.14** in the low-price scenario, **14.74** in the high-price scenario, and **14.52** in the random scenario. This confirms that the system performs best when heat recovery and scrap minimization are prioritized alongside renewable electricity use. The radar chart emphasizes the **multidimensional** circularity performance, as presented on Figures 12, 13, 14.

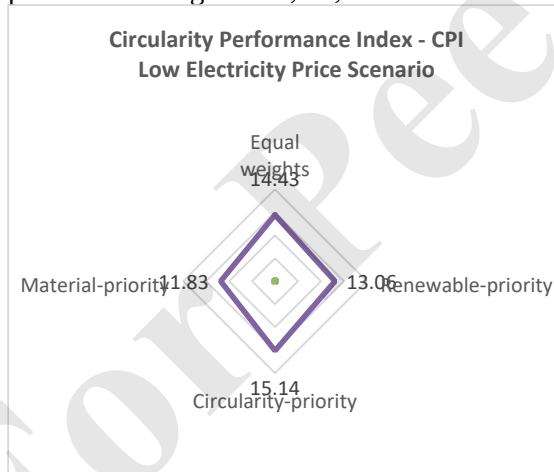


Figure 12.

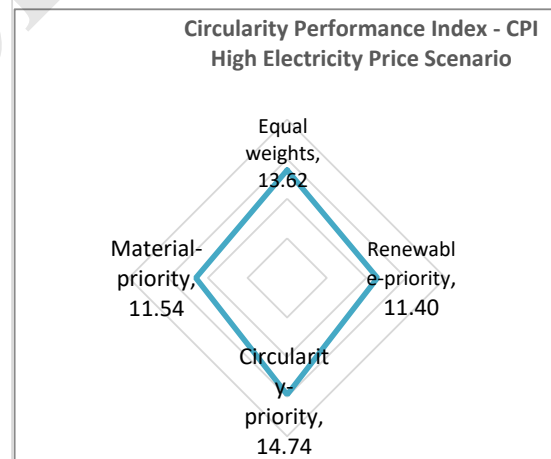


Figure 13.

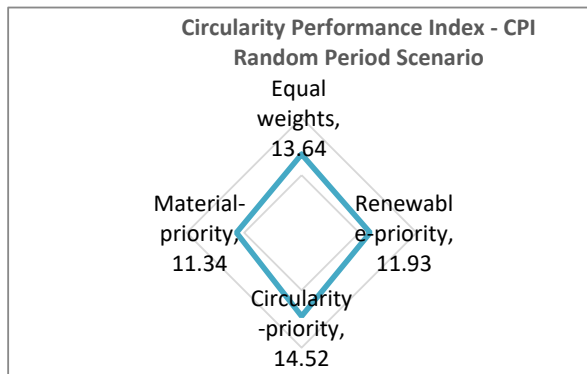


Figure 14.

Figures 12, 13, 14. CPI across Electricity - market scenarios

6. Discussion

The results reveal that operational flexibility, renewable integration, and thermal energy recirculation interact strongly with electricity-market conditions, significantly affecting both economic performance and operational circularity outcomes. Across all examined indicators, the **Low Electricity Price Scenario** consistently exhibits the most favorable balance between electricity cost minimization, renewable self-consumption, and circularity performance. This indicates that periods characterized by lower wholesale electricity prices are not necessarily detrimental to industrial prosumers; on the contrary, **they often coincide with increased renewable penetration and lower marginal generation costs, thereby creating favorable operating conditions for flexible industrial scheduling.** Under these conditions, the optimization model successfully aligns production tasks with photovoltaic generation availability and low-cost electricity hours, maximizing renewable electricity utilization and reducing grid dependence during economically advantageous periods. The significantly higher contribution of renewables to cost minimization in the low-price scenario confirms that the value of renewable integration depends not only on installed PV capacity, but also on the temporal synchronization between renewable generation, industrial load flexibility, and storage operation.

The battery-storage results further reinforce this interpretation. Although the standalone contribution of the battery remains smaller than the overall renewable contribution, the results indicate that battery value is highly sensitive to the PV-only charging constraint imposed in the model. Because storage can only charge from locally generated photovoltaic surplus, the battery acts primarily as a renewable-energy time-shifting mechanism rather than as a conventional electricity arbitrage device. Consequently, battery contribution declines significantly during the high-price scenario because reduced synchronization between solar generation and industrial operation limits the amount of renewable electricity available for charging. This finding highlights an important distinction between industrial prosumer storage systems and conventional market-based arbitrage models: under renewable-constrained charging strategies, **the economic value of storage depends fundamentally on renewable availability and operational flexibility rather than purely on electricity price spreads.** This reveals an **important dynamic** within industrial prosumer systems: **storage profitability is not solely a function of electricity-price volatility, but rather a function of the simultaneous interaction** between renewable production timing, operational flexibility windows, and process continuity constraints.

The thermal-buffer and recovered-heat results demonstrate that thermal circularity becomes particularly important under stressed electricity-market conditions. The relatively high RHU values observed across all scenarios indicate that **recovered process heat constitutes a stable**

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4 **and robust internal energy source within steel manufacturing systems.** Interestingly, the
5 highest RHU occurs during the high-price scenario, suggesting that the optimization algorithm
6 intensifies heat recovery and thermal reuse when electricity procurement becomes more
7 expensive. This behavior reflects **the role of thermal buffering as an internal hedging**
8 **mechanism against volatile electricity prices.** Rather than relying exclusively on electrical
9 flexibility, the model exploits the thermal inertia of reheating, finishing, drying, and
10 metallurgical support processes to reduce external energy demand. This confirms that thermal
11 storage and waste-heat recovery are not merely auxiliary energy-efficiency technologies, but
12 **central components of operational circularity** within energy-intensive industrial systems.
13 The environmental indicators reveal similarly important dynamics. The Renewable Electricity
14 Share decreases substantially during the high-price scenario, indicating that **elevated**
15 **electricity prices are associated with less favorable renewable operating conditions and**
16 **reduced synchronization between photovoltaic availability and industrial demand.** At the
17 same time, Electricity Emissions Intensity increases significantly under high-price conditions,
18 suggesting that **expensive electricity periods are also characterized by more carbon-**
19 **intensive electricity generation mixes,** likely due to increased dependence on natural-gas-
20 fired marginal generation. An important hidden implication of this finding is that **electricity-**
21 **price signals themselves may partially function as indirect carbon signals,** especially in
22 electricity systems where high prices coincide with fossil-based marginal generation.
23 Therefore, **flexible industrial scheduling may contribute to decarbonization even in the**
24 **absence of explicit carbon-price optimization.**

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26 One of the most important findings concerns the behavior of the **Scrap Generation Rate.**
27 Despite substantial operational flexibility and demand-response participation, scrap rates
28 remain **remarkably stable across all scenarios.** This demonstrates that the optimization
29 framework successfully preserves metallurgical and operational constraints while
30 implementing flexible scheduling strategies. The **low variability in scrap generation**
31 **validates the concept of “quality-preserving curtailment”** introduced in this study, **whereby**
32 **flexible operation is constrained to avoid excessive thermal instability, abrupt process**
33 **interruptions, and unrealistic load ramping.** This finding is particularly important because
34 many simplified industrial demand-response models implicitly assume that industrial
35 electricity demand can be shifted without affecting production quality. The present results
36 show that flexibility can indeed be achieved while maintaining process stability, provided that
37 operational constraints are explicitly integrated into the optimization problem.

38 The Circularity Performance Index further highlights the multidimensional nature of
39 operational circularity. **The CPI results demonstrate that operational circularity cannot be**
40 **reduced to a single variable such as renewable penetration or recycling intensity.** The
41 increase in CPI under weighting schemes emphasizing heat recovery and scrap minimization
42 confirms that **circularity in steel manufacturing cannot be evaluated solely through**
43 **renewable electricity penetration.** Instead, **internal energy recirculation and material-**
44 **efficiency preservation play equally important roles** in determining overall circularity
45 performance. The downstream rolling and finishing configuration **exhibits particularly**
46 **strong circularity performance** because its processes possess greater **temporal flexibility**
47 **and weaker metallurgical continuity constraints than EAF-dominated mini-mills.** This
48 confirms that **operational flexibility** itself constitutes an **important circularity-enabling**
49 **mechanism** within industrial systems.

50 Finally, the evolution of daily costs over the seven-day horizon reveals that industrial electricity
51 expenditures are driven not only by average electricity prices but also by the temporal
52 distribution and persistence of price spikes. The high-price scenario exhibits sustained elevated
53 daily costs throughout the entire week, while the random scenario displays isolated but extreme
54 cost peaks. This distinction suggests that industrial flexibility strategies must address both
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4 persistent market stress and short-term volatility. Overall, the findings demonstrate that
5 **operational circularity, renewable integration, and industrial flexibility should not be**
6 **viewed as independent concepts, but rather as deeply interconnected mechanisms capable**
7 **of simultaneously improving economic performance, reducing environmental impacts,**
8 **and preserving industrial process quality within future low-carbon electricity systems.**

9 The demand-function behavior highlights an important distinction between industrial demand-
10 response systems and conventional microeconomic demand theory. Although the resulting
11 curves preserve the fundamental inverse relationship between electricity price and electricity
12 demand, the underlying mechanism differs substantially from classical consumer-demand
13 behavior.
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15 In the present framework, industrial electricity demand is technologically constrained by
16 required production loads associated with EAF melting campaigns, downstream rolling
17 operations, reheating sequences, and metallurgical continuity requirements. Consequently, the
18 industrial plant does not increase electricity consumption simply because electricity prices
19 become low, nor can it arbitrarily reduce required electricity demand during expensive periods
20 without compromising production continuity and process quality. Instead, the optimization
21 framework reallocates fixed or semi-fixed production loads temporally across intraday
22 electricity-market periods in order to minimize operational cost while preserving metallurgical
23 stability and minimizing scrap generation.

24 The asymptotic form of the demand curves therefore reflects the gradual exhaustion of
25 economically feasible flexibility windows rather than pure consumption elasticity. The
26 pronounced curvature observed at low-price regions indicates that **industrial flexibility is**
27 **most valuable when electricity markets provide substantial low-cost renewable-rich**
28 **periods** that allow concentration of electricity-intensive process stages.

29 Conversely, the flattening of the curves at higher electricity prices reflects the persistence of
30 unavoidable technological demand associated with critical industrial operations. Another
31 important finding is that changes in total operational cost **shift the demand curves vertically**
32 across different production days and scenarios.

33 Higher operational costs tend to shift the curves upward and to the right, indicating increased
34 economically required electricity utilization under stressed operating conditions, whereas
35 lower-cost periods allow more efficient scheduling and lower effective demand levels for
36 equivalent electricity prices.

37 Overall, the results confirm that industrial demand response in EAF steel production should be
38 interpreted **primarily as a problem of temporal flexibility and operational scheduling**
39 **optimization** rather than conventional quantity-based demand elasticity.
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42 43 7. Limitations and Future Research Directions

44 Despite the strong techno-economic and operational contributions of the proposed framework,
45 several limitations should be acknowledged. First, the present optimization model is based on
46 a deterministic representation of electricity prices, photovoltaic generation, and industrial
47 operating conditions. Although the selected scenarios capture realistic periods of low-price,
48 high-price, and random electricity-market behavior, the framework does not explicitly model
49 uncertainty associated with renewable intermittency, electricity-market volatility, equipment
50 outages, or unexpected production disturbances. Future research could therefore extend the
51 methodology toward stochastic or robust optimization approaches capable of representing
52 uncertainty in electricity markets and renewable generation forecasts.

53 Second, the industrial production system is represented through a simplified process-level
54 decomposition of the Electric Arc Furnace (EAF) mini-mill and downstream rolling/finishing
55 configurations. Although the adopted process structure captures the principal operational
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characteristics of steel manufacturing, several detailed metallurgical phenomena—such as dynamic furnace chemistry, electrode consumption, refractory degradation, thermal transfer dynamics, and steel-grade-specific production constraints—are outside the scope of the current framework. Consequently, future work may incorporate higher-resolution thermodynamic and metallurgical process models in order to improve the physical realism of industrial flexibility representation.

Another limitation concerns the representation of battery storage and thermal buffering systems. The current framework is modeling battery efficiency and thermal storage operation using simplified linear relationships suitable for Mixed-Integer Linear Programming (MILP). However, real industrial storage systems exhibit nonlinear charging/discharging efficiencies, thermal losses, degradation mechanisms, and dynamic operational constraints. Similarly, the thermal buffer subsystem approximates industrial waste-heat recovery through aggregated thermal energy flows rather than detailed thermodynamic heat-exchanger behavior. Future studies could therefore integrate nonlinear electrochemical battery models, dynamic thermal-storage behavior, and more advanced heat-recovery network optimization approaches.

The present work also focuses primarily on short-term operational optimization over repeated 24-hour industrial process circles within a 7-day scheduling horizon. Although this structure realistically captures industrial demand-response behavior and intraday electricity-market participation, longer-term strategic decisions such as investment optimization, equipment retrofitting, renewable-capacity expansion, hydrogen integration, or long-term decarbonization pathways are not explicitly modeled. Future research may therefore combine the operational optimization framework with long-term capacity-expansion and investment-planning models in order to evaluate optimal industrial transition pathways under evolving energy-transition policies and carbon-pricing mechanisms.

Additionally, the environmental assessment is based mainly on operational emissions indicators associated with electricity consumption and natural gas use. While the framework introduces operational circularity indicators and process-level environmental metrics, it does not perform a fully integrated consequential Life Cycle Assessment (LCA) including upstream raw-material extraction, transportation, infrastructure construction, equipment manufacturing, or end-of-life recycling effects. Future work may therefore integrate dynamic LCA methodologies directly into the optimization objective function, enabling simultaneous optimization of operational costs, carbon emissions, and life-cycle environmental impacts.

Another important limitation concerns electricity-market participation mechanisms. The present study models industrial flexibility primarily through exposure to hourly electricity prices and self-consumption optimization. However, future electricity systems may increasingly include participation in ancillary-service markets, balancing markets, flexibility-capacity mechanisms, and local energy communities. Consequently, future research could extend the framework toward multi-market participation strategies where industrial prosumers simultaneously optimize electricity procurement, reserve provision, balancing participation, and flexibility-service revenues.

Finally, the proposed concept of operational circularity represents an emerging research direction that requires further theoretical and empirical development. Although the study introduces a multidimensional Circularity Performance Index combining renewable electricity utilization, recovered heat reuse, and scrap minimization, the weighting structure remains partially assumption-based and may vary according to industrial priorities, policy objectives, or regional circular-economy strategies. Future research could therefore explore data-driven weighting methodologies, entropy-based approaches, stakeholder-oriented circularity valuation, and alignment with evolving European Union circular economy taxonomies and industrial sustainability reporting frameworks.

Overall, despite these limitations, the proposed framework establishes a strong foundation for future research on industrial demand response, operational circularity, renewable integration, and low-carbon industrial optimization. The methodology demonstrates substantial potential for extension toward hydrogen-based steelmaking, carbon capture integration, sector-coupled industrial energy systems, digital-twin industrial optimization, and AI-driven industrial flexibility management under future renewable-dominated electricity markets.

8. Conclusions

The present study developed and evaluated an integrated techno-economic optimization framework for electricity-intensive steel manufacturing systems operating under dynamic electricity-market conditions and industrial demand response environments. The proposed methodology combines industrial production scheduling, renewable electricity integration, battery energy storage, thermal energy recovery, and operational circularity assessment within a unified Mixed-Integer Linear Programming (MILP) framework. Unlike conventional industrial optimization approaches that treat electricity demand as fixed and exogenous, the present work explicitly integrates the production-sequencing logic of Electric Arc Furnace (EAF)-based steel manufacturing into the optimization process, enabling industrial operations to dynamically respond to electricity-market signals, renewable availability, and thermal recovery opportunities. In doing so, the study contributes to bridging the gap between industrial process engineering, energy economics, demand response, and circular economy research.

The results demonstrate that operational flexibility can significantly improve the economic and environmental performance of steel-sector industrial prosumers. The integration of photovoltaic generation, battery storage, thermal buffering, and process-level scheduling flexibility reduces electricity procurement costs while simultaneously increasing renewable self-consumption and internal energy recirculation. The Low Electricity Price Scenario consistently produced the most favorable operational outcomes, including the lowest total operational cost, the highest renewable contribution to cost minimization, and the strongest Circularity Performance Index (CPI) values. This finding indicates that industrial flexibility becomes particularly valuable under electricity-market conditions characterized by high renewable penetration and lower marginal electricity prices. The optimization framework successfully aligns electricity-intensive process stages with favorable electricity-market periods, allowing the industrial facility to reduce exposure to expensive or carbon-intensive electricity hours while maintaining realistic operational continuity constraints.

A major contribution of the study concerns the explicit modeling of operational circularity. Existing circular economy studies in the steel sector focus primarily on material recycling, slag valorization, and by-product recovery. In contrast, the present work extends circularity toward an operational and energy-management perspective, introducing the concept of operational circularity as the coordinated interaction between renewable electricity utilization, thermal energy reuse, process-quality preservation, and flexible industrial scheduling. The findings demonstrate that industrial circularity can be improved not only through material recovery processes, but also through intelligent synchronization of production operations with renewable electricity generation and waste-heat recovery availability. In particular, the thermal buffer subsystem proved to be a critical mechanism for internal energy recirculation, reducing dependence on external electricity and fuel consumption by reusing recoverable heat from metallurgical and downstream thermal processes. The consistently high Recovered Heat Utilization (RHU) values across all scenarios highlight the strategic importance of thermal energy integration in future low-carbon industrial systems.

The battery-storage subsystem also provided important insights regarding the role of renewable-constrained industrial storage operation. Under the imposed PV-only charging constraint, the battery acted primarily as a renewable-energy temporal balancing mechanism

rather than as a conventional electricity-market arbitrage device. Consequently, the economic contribution of battery storage depended strongly on the synchronization between photovoltaic availability, industrial scheduling flexibility, and electricity-price variability. This finding suggests that future industrial prosumer systems may derive greater value from coordinated renewable-storage scheduling rather than from purely market-driven electricity arbitrage strategies. Similarly, the results showed that electricity-price signals indirectly influence industrial emissions intensity because periods characterized by elevated electricity prices are often associated with fossil-fuel-dominated marginal electricity generation. Therefore, industrial demand response can simultaneously reduce operational costs and indirect carbon emissions by shifting electricity-intensive operations away from carbon-intensive market periods.

Another particularly important finding concerns the stability of scrap-generation rates across all examined scenarios. Despite the implementation of extensive operational flexibility and load-shifting strategies, scrap rates remained remarkably close to baseline levels. This validates the concept of “quality-preserving curtailment” introduced in the study, whereby operational flexibility is constrained within realistic metallurgical and process-stability boundaries. The optimization framework therefore avoids unrealistic over-flexibilization that could otherwise lead to thermal instability, process discontinuities, reheating losses, or material-quality degradation. This result is especially significant because many simplified industrial demand-response models implicitly assume unrestricted flexibility without accounting for the operational realities of industrial process continuity and product-quality preservation. The proposed framework demonstrates that meaningful industrial flexibility can be achieved while simultaneously preserving process reliability and minimizing material losses.

The comparative analysis between the mini-mill configuration and downstream rolling/finishing plants further revealed that industrial flexibility potential depends not only on electricity intensity but also on the technological rigidity of the production sequence. Downstream rolling and finishing facilities exhibited greater scheduling flexibility and higher operational circularity potential because their production stages can be shifted within broader operational windows without violating strict metallurgical continuity constraints. In contrast, EAF-dominated mini-mills remain more constrained due to the discrete and thermally intensive nature of furnace heat cycles. This distinction suggests that future industrial demand-response programs and flexibility markets may need to differentiate industrial sectors according to their operational and thermal characteristics rather than treating industrial demand as homogeneous flexible load.

Overall, the developed framework demonstrates that modern steel-production systems can evolve from passive electricity consumers into actively optimized industrial prosumers capable of participating dynamically in future renewable-rich electricity systems. By integrating industrial scheduling, renewable electricity utilization, thermal recovery, storage operation, emissions reduction, and circularity assessment into a unified optimization architecture, the study provides a scalable and realistic techno-economic framework for evaluating industrial flexibility and decarbonization pathways under energy-transition conditions. The methodology is fully compatible with future extensions involving green hydrogen integration, low-carbon synthetic fuels, advanced thermal-storage systems, and sector-coupled industrial-energy infrastructures. Consequently, the framework offers a valuable analytical and policy-oriented tool for supporting the transition of electricity-intensive industries toward more flexible, circular, and low-carbon industrial production systems.

The demand-function analysis confirms that electricity demand in EAF-based steel production behaves fundamentally differently from classical price-elastic consumer demand. Industrial electricity consumption remains technologically constrained and largely inelastic in quantity, while flexibility is expressed mainly through the temporal scheduling of required production

loads across electricity-market periods. The asymptotic demand behavior demonstrates that low-price periods strongly attract electricity-intensive operations, whereas high-price conditions cannot fully suppress industrial demand because minimum production continuity must be preserved. Consequently, industrial demand response within renewable-rich electricity systems should be interpreted as a process-scheduling optimization mechanism that improves economic efficiency, renewable-energy utilization, and operational circularity without compromising metallurgical stability or production continuity.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

The data that support the findings of this study are available from the corresponding author Efstathios (**Stathis**) **Devves**, in the restored Zenodo Github link. <https://doi.org/10.5281/zenodo.20597099>. In this repository link main codes, data set and output files are included into the corresponding folders.

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